MAST20026 Real Analysis Semester 2 2025

Assignment 2

Due: 28 August, 11:59pm

M= a mark associated with applying an appropriate method

J= a mark associated with using correct justification

A= a mark associated with a correct answer or intermediate step

N= a mark associate with good notation or writing

Please see the assignment instructions on the Assignment 2 page on the LMS, where you accessed this file. To ease submission, please prepare assignment solutions for each question on its own page. This document is 2 pages in length.

PART A (*2 marks* total)

Go to the article *Why Mathematical Proof Is a Social Compact* linked from the Assignment 2 page on the LMS. Read it and write a short reflective response (around half a page) that considers the following questions.

- Was there anything about the reading that you found surprising?
- How does Andrew Granville's notion of a mathematical proof align with the notions of proof discussed in this subject, and your own thoughts on what a proof is?
- The article argues that mathematical proof is inherently social and built on shared understanding. How do you think this plays out when there are many different social groups and cultures around the world?

1 mark for demonstrating thoughtful engagement with the task.
1 mark for clearly communicated response (the response is well written, and conveys thoughts concisely).

PART B (*22 marks* total)

1. **Preimages and images**. (*10 marks*) Let X and Y be sets, and let $f : X \to Y$ be a function. Recall that given a function $f : X \to Y$, and a subset $S \subseteq Y$, the *preimage* of S is the set

$$f^{-1}(S) = \{x \in X \mid f(x) \in S\}.$$

   (a) Provide a (semi)-formal proof that for all $A, B \subseteq Y$,

$$\text{if } A \subseteq B \text{ then } f^{-1}(A) \subseteq f^{-1}(B).$$

   (b) Recall that for a subset $R \subseteq X$, the *image* of R is

$$f(R) = \{f(x) \mid x \in R\}.$$

1M: A demonstrated understanding of what a (semi-)formal proof is.

1M: Demonstrated understanding of how a correct proof of the statement should proceed

i. Is it true that $f^{-1}(f(C)) = C$ for all subsets C of X? If it is, provide a proof. If not, provide a counter-example.

ii. Is it true that $f(f^{-1}(D)) = D$ for all subsets D of Y? If it is, provide a proof. If not, provide a counter-example.

Solution.

1J: The statements / conditions in the lines of the proof follow a logical order and are mostly correct. (At most 2 minor errors)

1J: The justifications for each line in the proof mostly reference the correct previous lines, and definitions, and explain the current line. (At most 2 minor errors)

1J: A complete and correct (semi-)formal proof, with work written out clearly.

(a) *Proof.*

1.	Let X and Y sets and let $f : X \to Y$ be a function.	(premise)
2.	Let $A, B \subseteq Y$ and $A \subseteq B$	(premise)
3.	Let $x \in X$ such that $x \in f^{-1}(A)$	(premise)
4.	$f(x) \in A$	(3., defn of preimage)
5.	$f(x) \in B$	(2., 4., defn of $\subseteq$)
6.	$x \in f^{-1}(B)$	(5. defn of preimage)
7.	$(\forall x \in X)\ [x \in f^{-1}(A) \implies x \in f^{-1}(B)]$	(3., 6.)
8.	$f^{-1}(A) \subseteq f^{-1}(B)$.	(7., defn of $\subseteq$)
9.	For all $f : X \to Y$ and all $A, B \subseteq Y$, $f^{-1}(A) \subseteq f^{-1}(B)$.	(1.,2.,8.)

For b) either in part i or ii
1M: Demonstrated understanding of image and pre-image (eg. not confusing pre-image with inverse function)

(b) i. Not True.. 1A: Not true

An f that is not injective would provide a counter-example.
For example let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$.
Consider $C = \{2\} \subseteq \mathbb{R}$.
Then $f(C) = \{4\}$.
But

1J: Any correct counter-example with valid reasoning

$$f^{-1}(f(C)) = f^{-1}(\{4\}) = \{-2, 2\} \neq \{2\} = C.$$

ii. Not True. 1A: Not true

An f whose image is a strict subset of D would provide a counter-example.
For example, let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$.
Consider $D = \mathbb{R} \subseteq \mathbb{R}$.
Then $f^{-1}(D) = \mathbb{R}$.
But

1J: Any correct counter-example with valid reasoning

$$f(f^{-1}(D)) = f(\mathbb{R}) = [0, \infty) \neq \mathbb{R} = D$$

2. **The identities.** (*5 marks*) Using only the Axioms of the Real Numbers (and the rules of logic), write a (semi-)formal proof of the following statement.

$$0 < 1$$

Here, 0 is the additive identity and 1 is the multiplicative identity, whose existences are guaranteed by the axioms.

1M: A demonstrated understanding of what a (semi-)formal proof using the axioms is.

1M: Demonstrated understanding of how a correct proof of the statement should proceed

1J: The lines of the proof follow a logical order and are mostly correct. (At most 3 minor errors)

1J: The justifications for each line in the proof mostly reference the correct previous lines and correct axioms of R (At most 3 minor errors)

1J: A complete and correct (semi-)formal proof using the real number axioms with work written out clearly. (At most 1 minor error)

Solution.
There are a few ways to do this. Here's one idea: If $1 < 0$ can show $-1 > 0$. Then deduce that $(1)(-1) < (0)(-1) \implies -1 < 0$, to get a contradiction.

We will need to know the following claim:

$$\text{Claim: } (0)(-1) = 0$$

Proof.

1.	$0 + 0 = 0$	(A4)
2.	$(-1)(0 + 0) = (-1)(0)$	(1., multiplication is well defined)
3.	$(-1)(0) + (-1)(0) = (-1)(0)$	(2., D)
4.	$((-1)(0) + (-1)(0)) + (-(-1)(0)) = (-1)(0) + (-(-1)(0))$	(3., A5, addition is well defined)
5.	$(-1)(0) + ((-1)(0) + (-(-1)(0))) = (-1)(0) + (-(-1)(0))$	(4., A3)
6.	$(-1)(0) + 0 = 0$	(5., A5)
7.	$(-1)(0) = 0$	(6., A4)

We know prove that

$$0 < 1$$

Proof. The proof will proceed by , showing $0 \neq 1$ and $1 \not< 0$.

1.	$0 \neq 1$	(M4)
2.	Let $1 < 0$	(premise, for contradiction)
3.	$1 + (-1) < 0 + (-1)$	(2. A5, OA)
4.	$0 < 0 + (-1)$	(3., A5)
5.	$0 < (-1) + 0$	(4., A2)
6.	$0 < -1$	(5. A4)
7.	$1(-1) < (0)(-1)$	(2., 6.,. OM)
8.	$-1 < 0$	(7., Claim above that $(0)(-1) = 0$)
9.	$0 < -1$ and $-1 < 0$ contradicts O1	(6., 8., O1)
10.	$1 < 0$ is false	(2., 9.)
11.	$0 < 1$	(1., 10. A1)

3. **Bounding sequences.** (*7 marks*) For $n \in \mathbb{N}$, let

$$a_n = \begin{cases} 1 & n = 0 \\ 1 & n = 1 \\ \frac{1}{a_{n-1}} + \frac{1}{a_{n-2}} & n \geq 2 \end{cases}$$

and let $S = \{a_n \mid n \in \mathbb{N}\}$.

(a) Use induction (or strong induction) to prove that 2 is an upper bound for S in $\mathbb{R}$.

(b) Does the set S have an infimum in $\mathbb{R}$? Justify your answer.

Solution.

1M: A demonstrated understanding of what a proof by induction/ strong induction.

1M: Demonstrated understanding of how a correct proof of the statement should proceed.

1J: Base case(s) correctly checked

1J: Induction step is mostly correct, and it is clear when the induction hypothesis is being used.

1J: Proof is correct and clearly written

(a) To show that 2 is an upper bound of S in $\mathbb{R}$, we will show that

$$a_n \leq 2, \text{for all } n \in \mathbb{N}.$$

It will be convenient to prove the stronger statement:

$$1 \leq a_n \leq 2, \text{for all } n \in \mathbb{N}$$

The proof will proceed by strong induction.
Let $p(n)$ be the condition that $1 \leq a_n \leq 2$.

Base case(s): We will prove $p(0)$ and $p(1)$.
By definition, $a_0 = a_1 = 1$.
Since $1 \leq 1 \leq 2$, $p(0)$ and $p(1)$ is true.

Inductive step: Prove $\forall k \in \mathbb{N}\ (p(0) \wedge p(1) \wedge \cdots p(k) \implies p(k+1))$.
We have

$$\begin{aligned} a_{k+1} &= \frac{1}{a_{k-1}} + \frac{1}{a_{k-2}} \\ &\leq \frac{1}{1} + \frac{1}{1} \quad (\ 1 \leq a_{k-1} \leq 2,\ 1 \leq a_k \leq 2, \text{ induction hypothesis}) \\ &= 2 \end{aligned}$$

Moreover

$$\begin{aligned} a_{k+1} &= \frac{1}{a_{k-1}} + \frac{1}{a_{k-2}} \\ &\geq \frac{1}{2} + \frac{1}{2} \quad (\ 1 \leq a_{k-1} \leq 2,\ 1 \leq a_k \leq 2, \text{ induction hypothesis}) \\ &= 1 \end{aligned}$$

By strong mathematical induction $1 \leq a_n \leq 2$ for all $n \in \mathbb{N}$.
Therefore for all S is bounded above by 2.

(b) Yes, S has an infimum.
We have shown 1 is a lower bound.
There can't be a lower bound that's greater since $1 \in S$.

1A: Yes S has an infimum.
1J: A correct reason is provided (eg. some recognition that S obtains the lower bound of 1.)